



Antennas: Principle and Design

**C. A. Balanis, “Antenna Theory Analysis and Design,”
3rd ed., New York, NY, USA: Wiley, 1997.**

Introduction



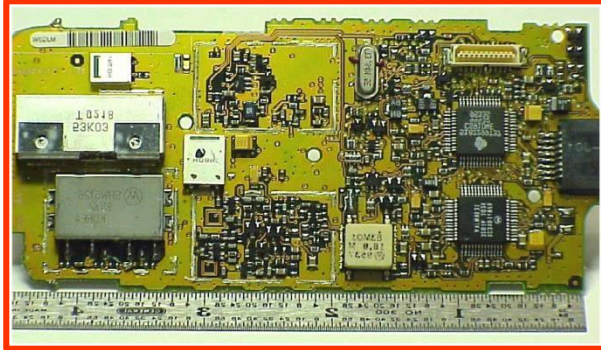
IEEE Definition of an antenna

- Antenna is that part of a transmitting or receiving system which is designed to radiate or to receive EM waves.

Proper design of this part is necessary for efficient coupling

Introduction

Two-way wireless transmission



Cellular phone handset

Key component: Antenna

Antenna:

as an interface

as an impedance transformer



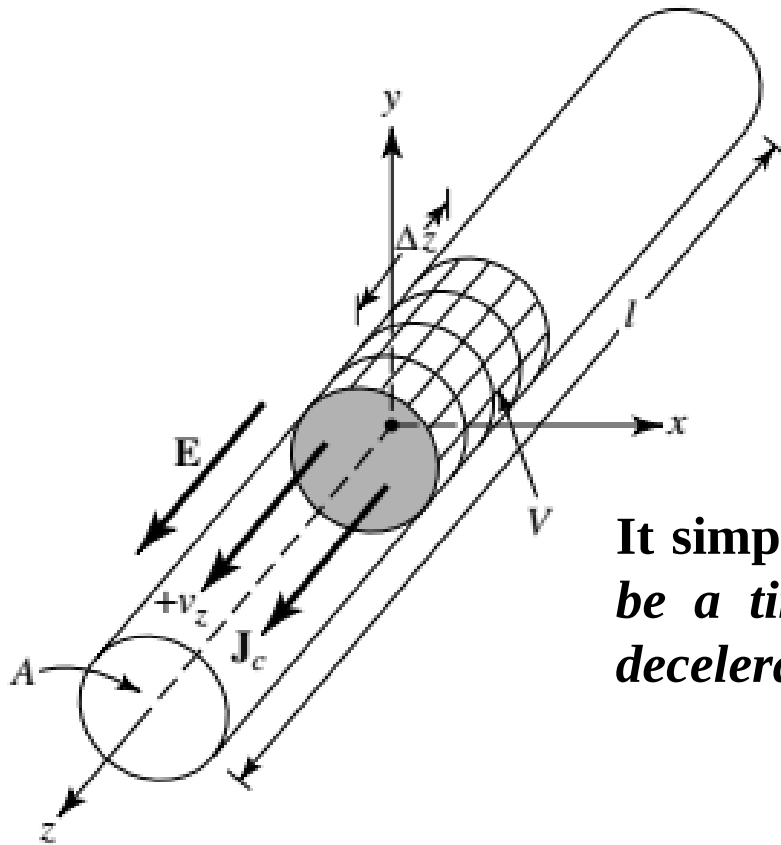
Base station Tower

Introduction



$$l \frac{dI_z}{dt} = l q_l \frac{dv_z}{dt} = l q_l a_z$$

Fundamental relation of electromagnetic radiation.



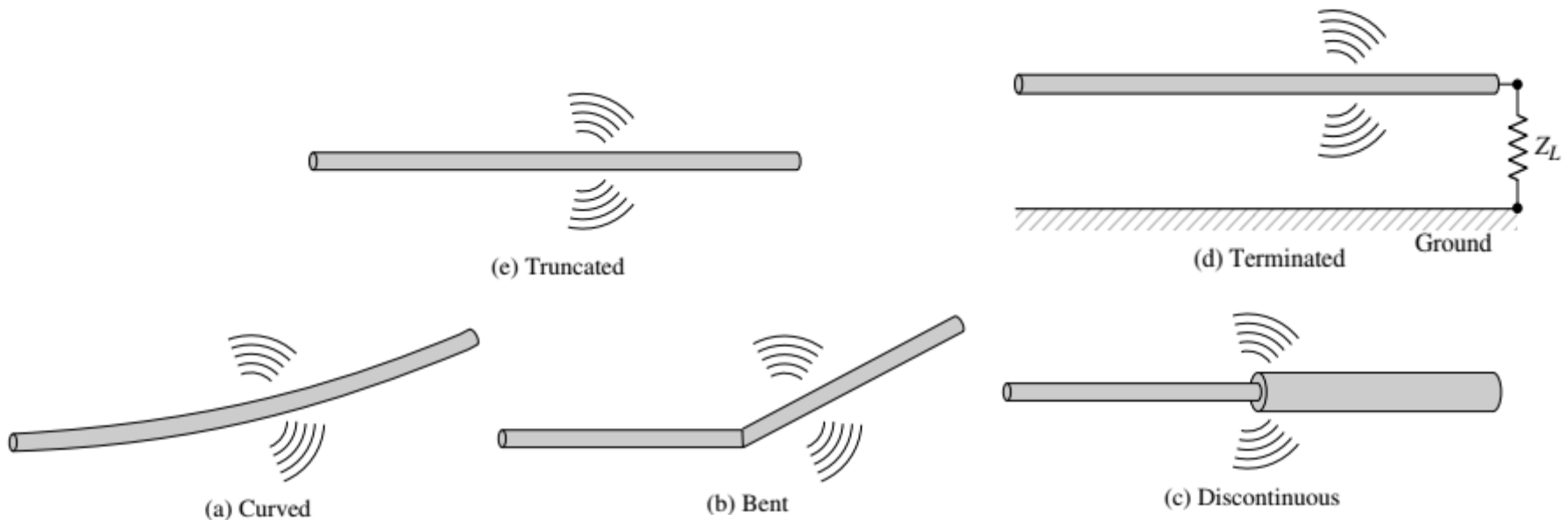
It simply states that to create radiation, there must be a time-varying current or an acceleration (or deceleration) of charge.

A charge is uniformly distributed in a circular cross-section cylinder wire.

Introduction



We typically refer to currents in time-harmonic applications, while charge is most often mentioned in transient applications. To create charge acceleration (or deceleration), the wire must be curved, bent, discontinuous, or terminated. Periodic charge acceleration (or deceleration) or time-varying current is also created when charge is oscillating in a time-harmonic motion



TYPES OF ANTENNAS

1. Wire Antennas: They are seen virtually everywhere—on automobiles, buildings, ships, aircraft, spacecraft, and so on. There are various shapes of wire antennas, such as a straight wire (dipole), a loop, and a helix

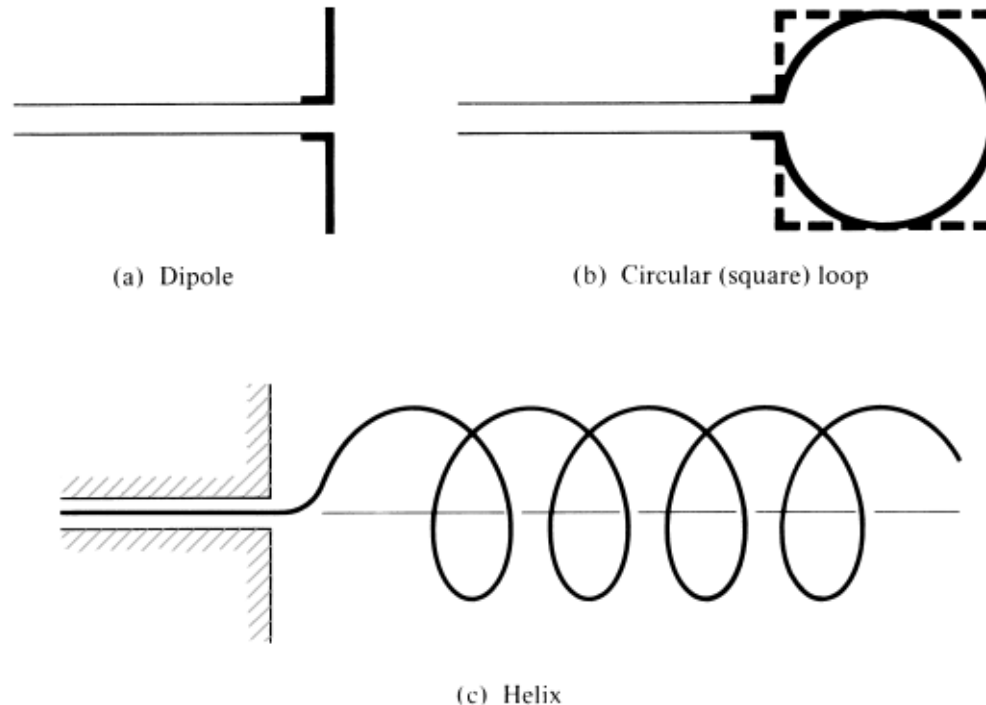
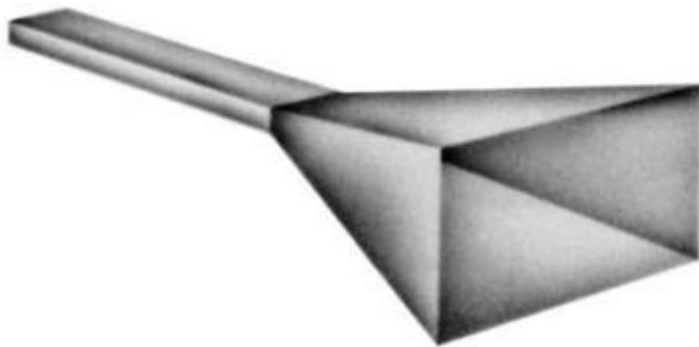


Figure 1.3 Wire antenna configurations.

TYPES OF ANTENNAS

2. Aperture Antennas

Antennas working at higher frequencies. Antennas of this type are particularly useful for aircraft and spacecraft applications, as they can be conveniently flush-mounted on the aircraft or spacecraft's skin. Additionally, they can be protected with a dielectric material to shield them from the hazardous environmental conditions.



(a) Pyramidal horn

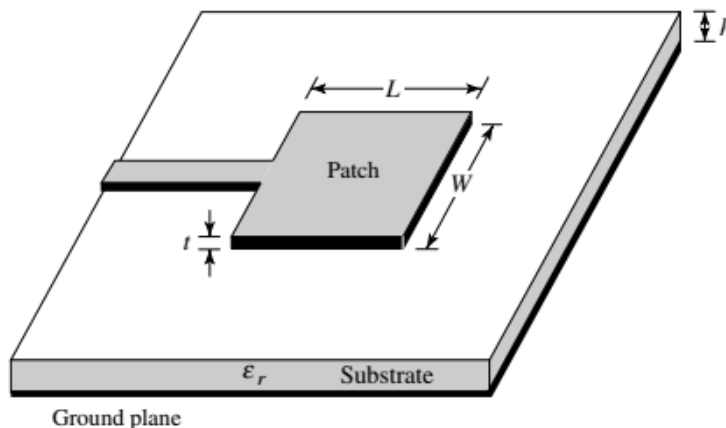


(b) Conical horn

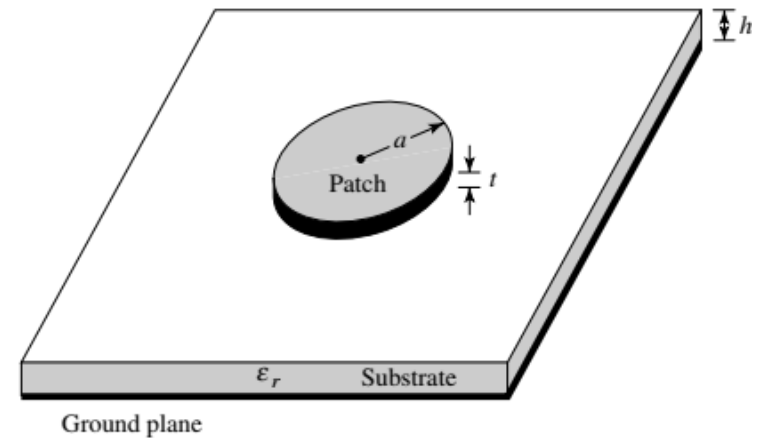
TYPES OF ANTENNAS

3. Microstrip Antennas

Microstrip antennas became very popular for space-borne applications. Today they are used for government and commercial applications. These antennas consist of a metallic patch on a grounded substrate. The metallic patch can take many different configurations, as shown below. However, the rectangular and circular patches, are the most popular because of ease of analysis and fabrication, and their attractive radiation characteristics, especially low cross-polarization radiation. The microstrip antennas are low-profile, conformable to planar and non-planar surfaces, simple and inexpensive to fabricate using modern printed-circuit technology, mechanically robust when mounted on rigid surfaces, compatible with MMIC designs, and very versatile in terms of resonant frequency, polarization, pattern, and impedance.



(a) Rectangular

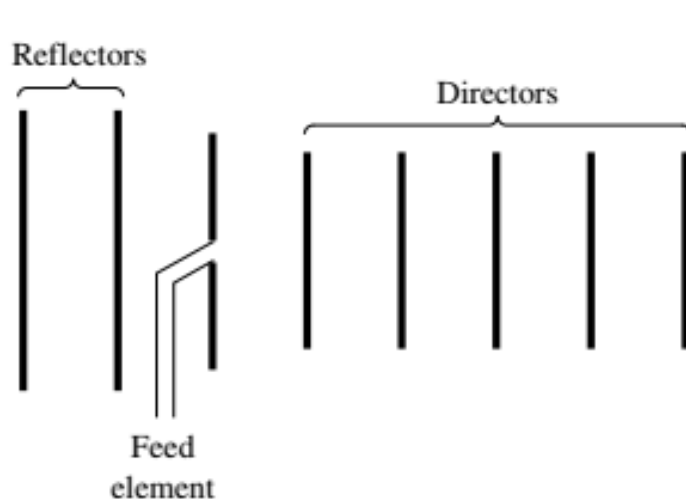


(b) Circular

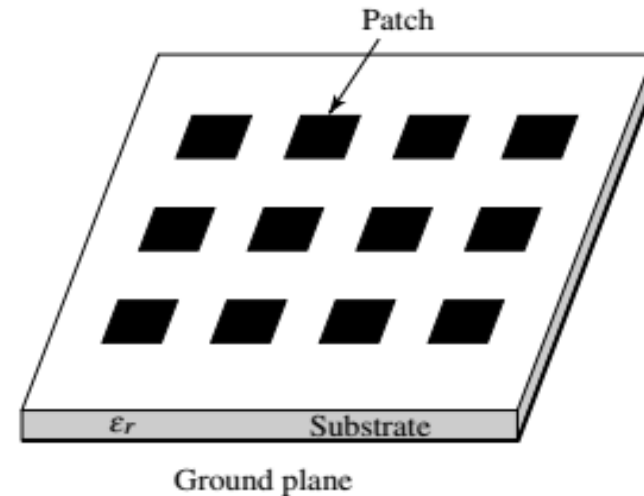
TYPES OF ANTENNAS

4. Array Antennas

Many applications require radiation characteristics that may not be achievable by a single element. It may, however, be possible that an aggregate of radiating elements in an electrical and geometrical arrangement (*an array*) will result in the desired radiation characteristics. The arrangement of the array may be such that the radiation from the elements adds up to give a radiation maximum in a particular direction or directions, a minimum in others, or otherwise as desired.



(a) Yagi-Uda array



(c) Microstrip patch array

RADIATION PATTERN

- An antenna *radiation pattern* or *antenna pattern* is defined as “a mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates.
- In most cases, the radiation pattern is determined in the far field region and is represented as a function of the directional coordinates.
- Radiation properties include power flux density, radiation intensity, field strength, directivity, and phase or polarization.
- A trace of the received electric (magnetic) field at a constant radius is called the amplitude field *pattern*. On the other hand, a graph of the spatial variation of the power density along a constant radius is called an amplitude *power pattern*.
 - a. *field pattern (in linear scale)* typically represents a plot of the magnitude of the electric or magnetic field as a function of the angular space.
 - b. *power pattern (in linear scale)* typically represents a plot of the square of the magnitude of the electric or magnetic field as a function of the angular space.
 - c. *power pattern (in dB)* represents the magnitude of the electric or magnetic field, in decibels, as a function of the angular space.

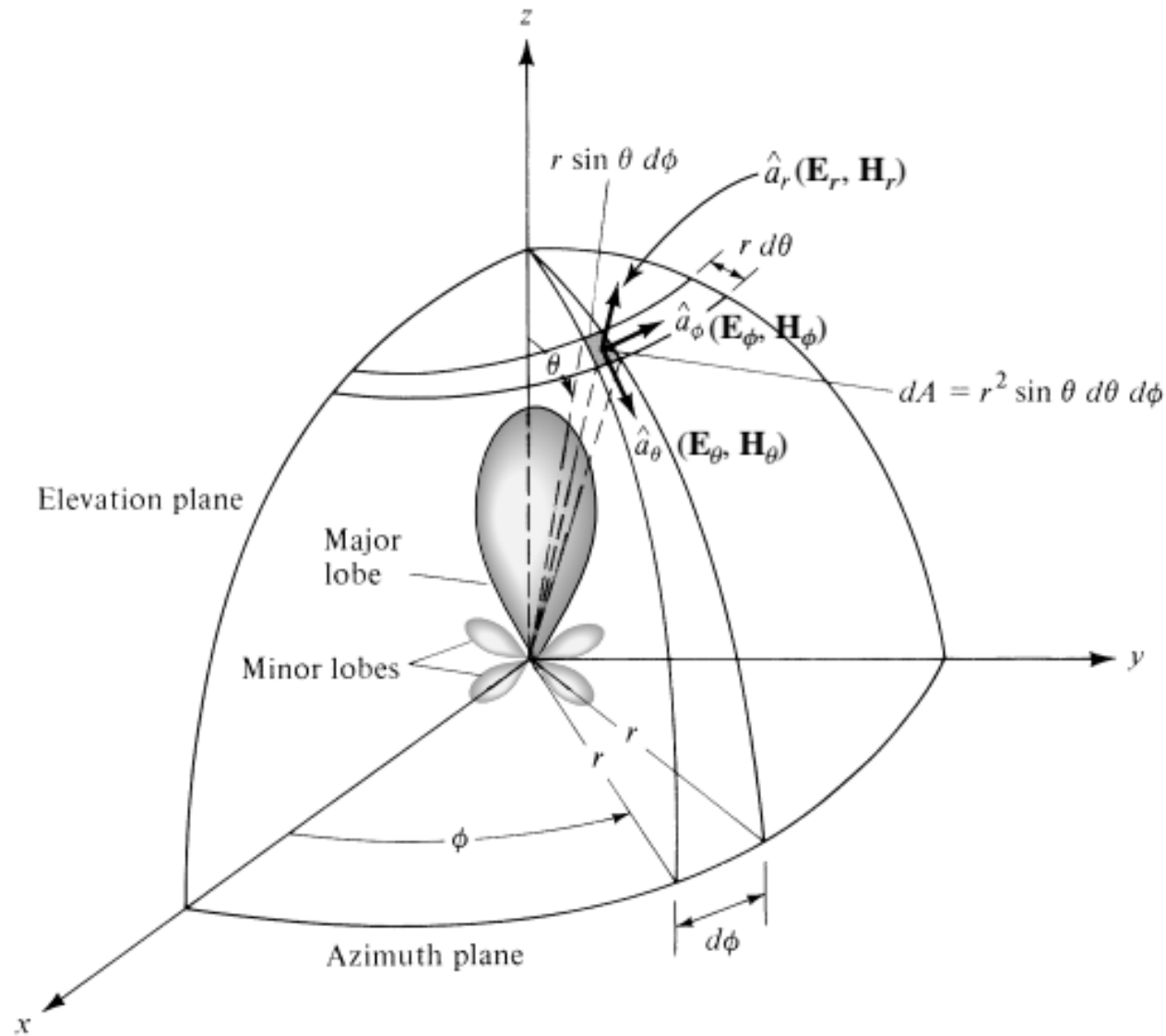
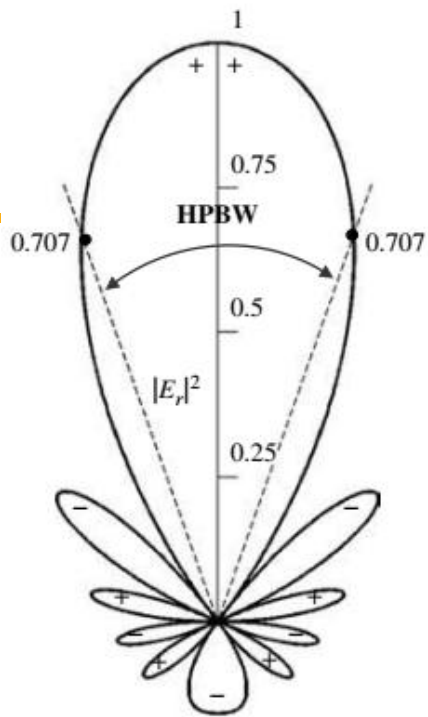
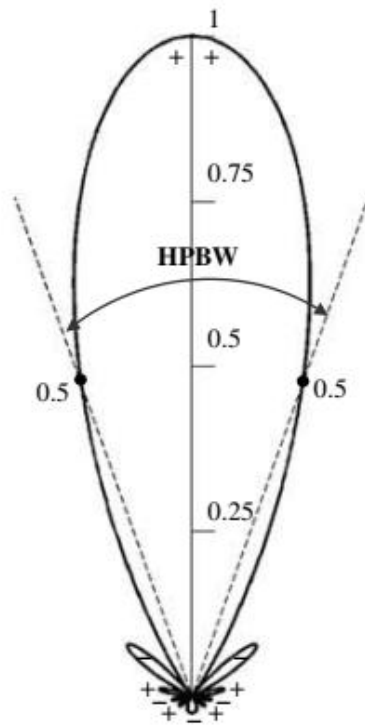


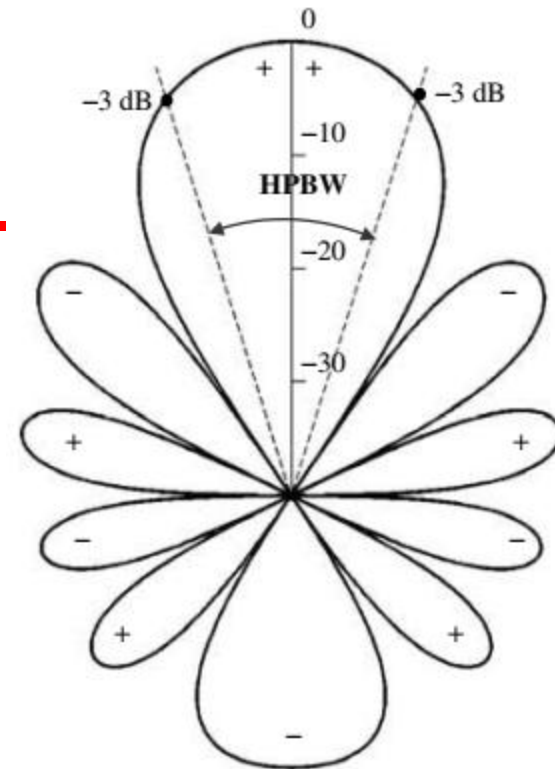
Figure 2.1 Coordinate system for antenna analysis.



(a) Field pattern (in linear scale)



(b) Power pattern (in linear scale)



(c) Power pattern (in dB)

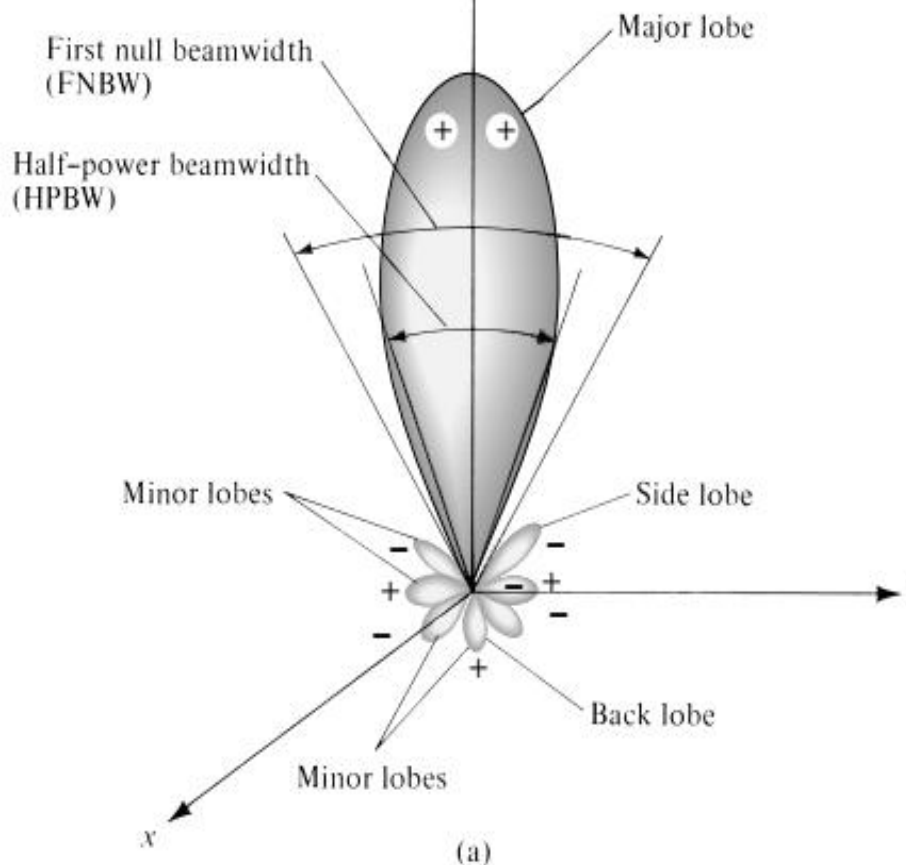
The two-dimensional normalized field pattern (*plotted in linear scale*), power pattern(*plotted in linear scale*), and power pattern (*plotted on a logarithmic dB scale*)

- field pattern at 0.707 value of its maximum, as shown in Figure 2.2(a)
- power pattern (in a linear scale) at its 0.5 value of its maximum, as shown in Figure 2.2(b)
- power pattern (in dB) at -3 dB value of its maximum, as shown in Figure 2.2(c).

Radiation Pattern Lobes



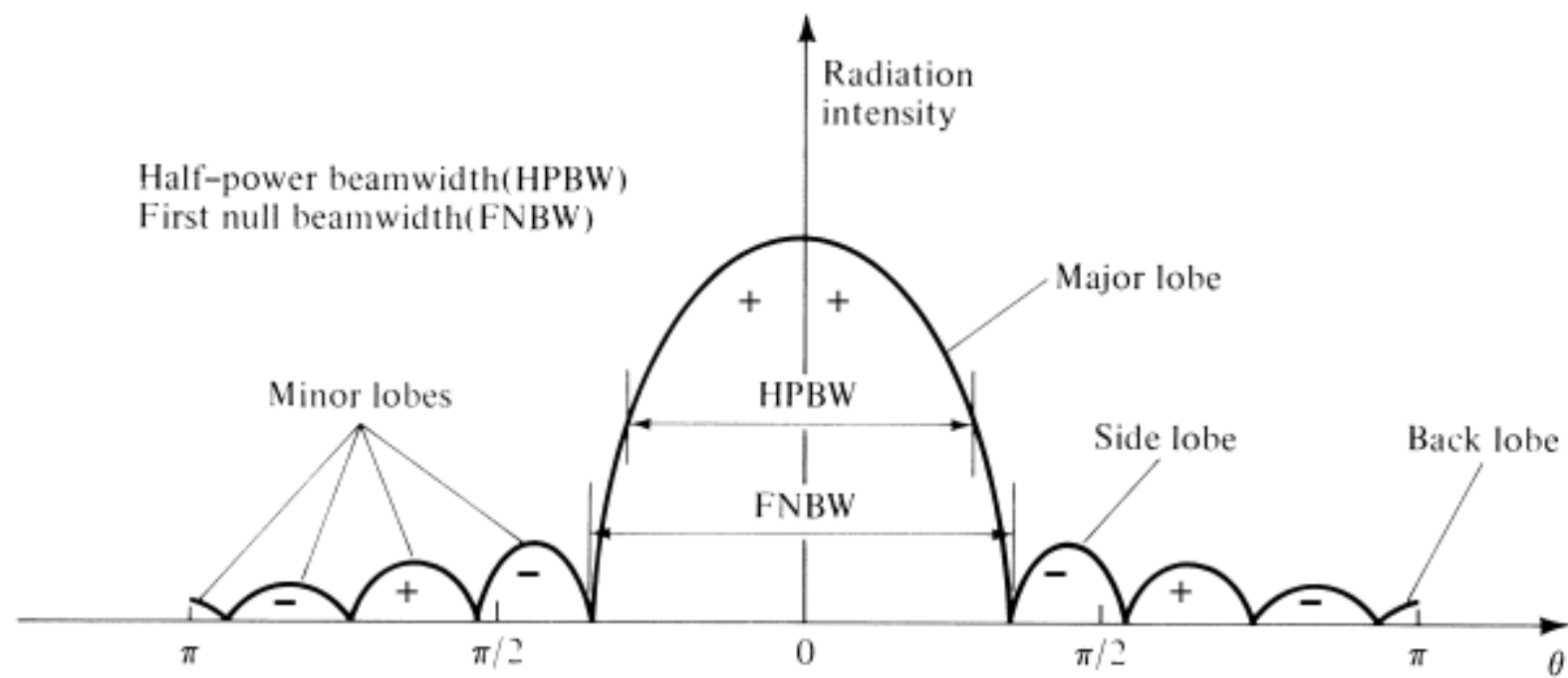
Various parts of a radiation pattern are referred to as *lobes*, which may be sub-classified into *major* or *main*, *minor*, *side*, and *back* lobes.



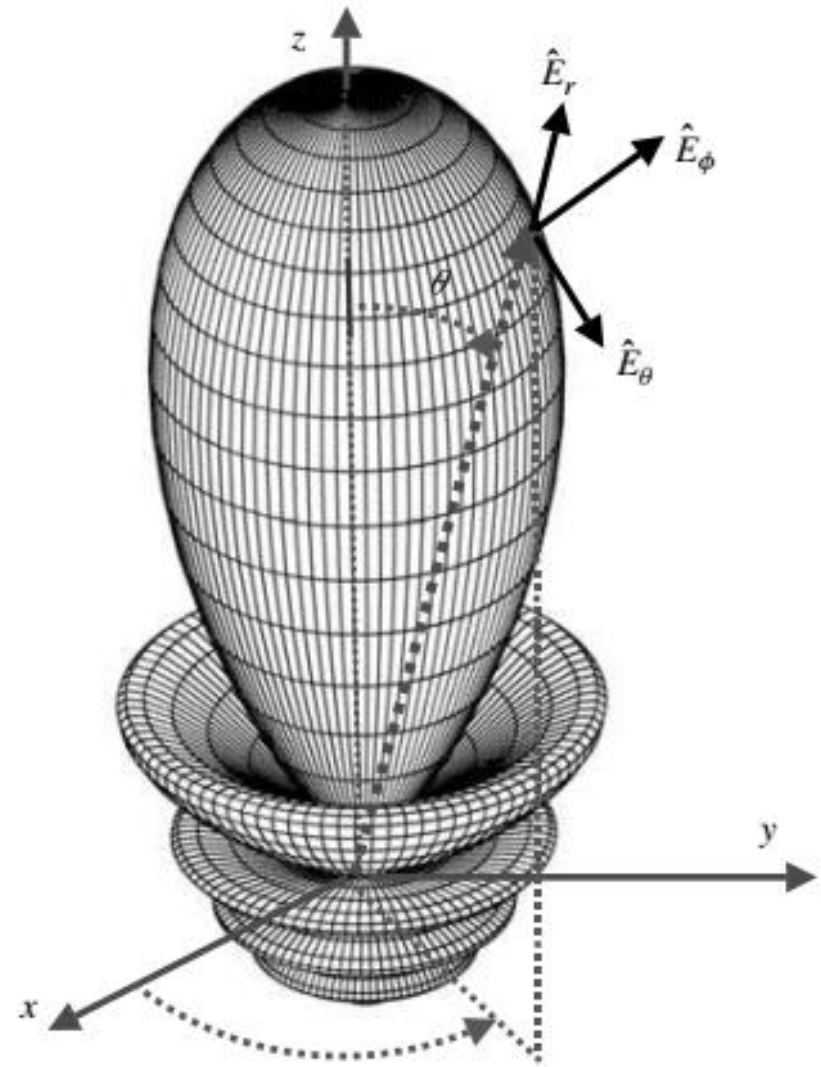
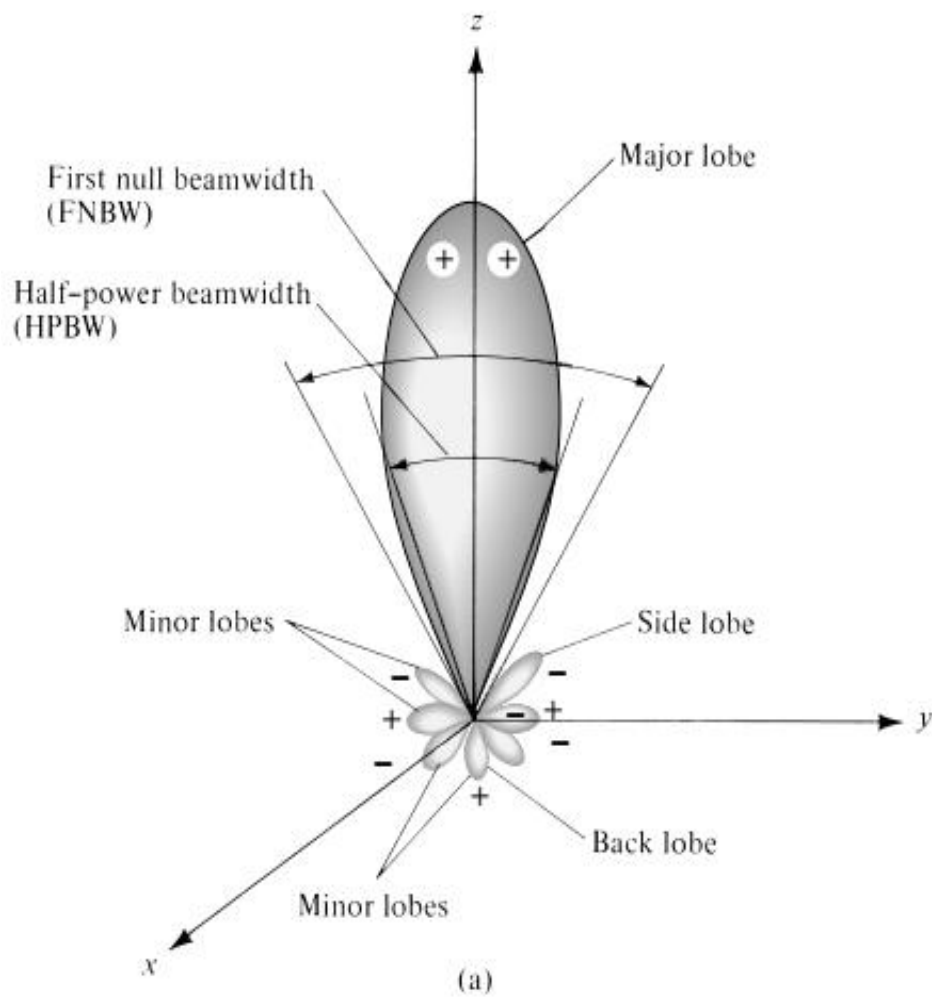
A major lobe (also called main beam) is defined as “the radiation lobe containing the direction of maximum radiation.” In Figure the major lobe is pointing in the $\theta = 0$ direction.

A *minor lobe* is any lobe except a major lobe. In Figure, all the lobes with the exception of the major can be classified as minor lobes. A *side lobe* is “a radiation lobe in any direction other than the intended lobe.” (Usually a side lobe is adjacent to the main lobe and occupies the hemisphere in the direction of the main beam.)

A *back lobe* is “a radiation lobe whose axis makes an angle of approximately 180° with respect to the beam of an antenna.” Usually it refers to a minor lobe that occupies the hemisphere in a direction opposite to that of the major (main) lobe.



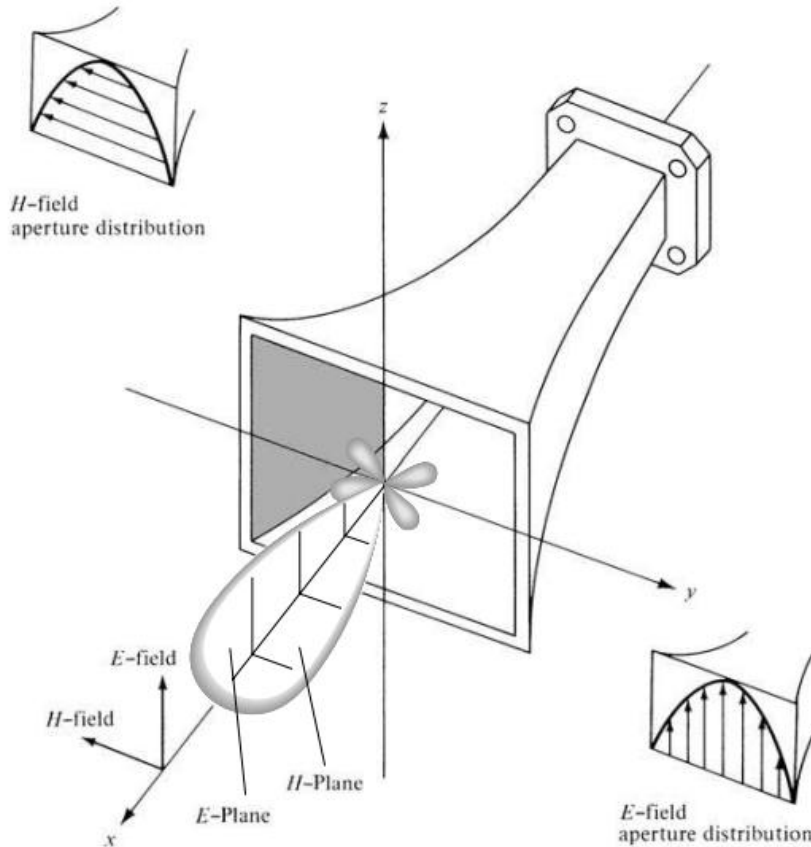
(b)



Isotropic, Directional, and Omnidirectional Patterns

An *isotropic* radiator is defined as “a hypothetical lossless antenna having equal radiation in all directions.” Although it is ideal and not physically realizable, it is often taken as a reference for expressing the directive properties of actual antennas.

Isotropic, Directional, and Omnidirectional Patterns

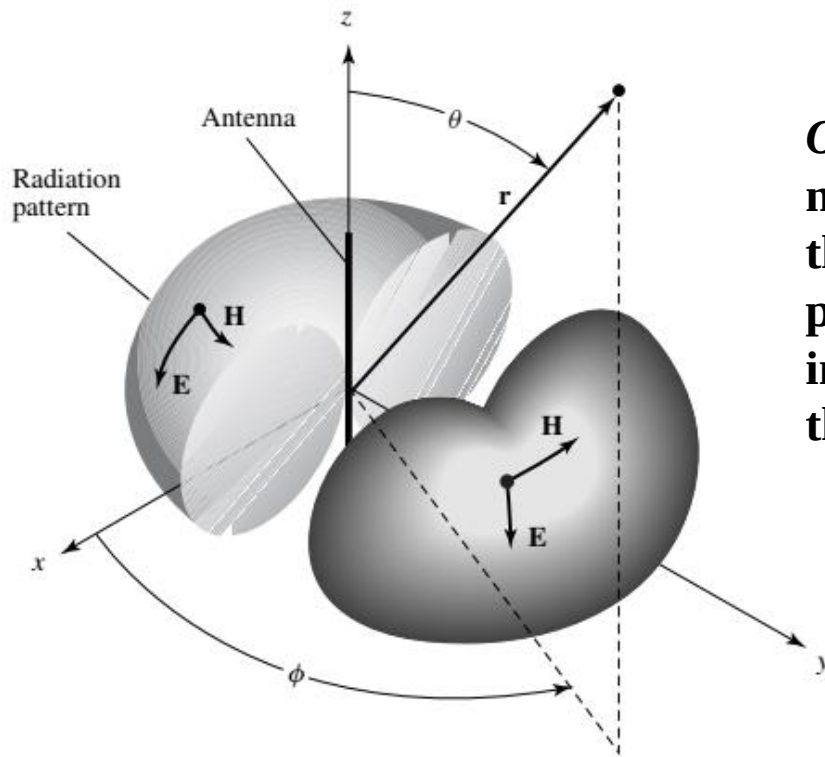


A

- A *directional* antenna is one “having the property of radiating or receiving electromagnetic waves more effectively in some directions than in others. This term is usually applied to an antenna whose maximum directivity is significantly greater than that of a half-wave dipole.”

B

Isotropic, Directional, and Omnidirectional Patterns

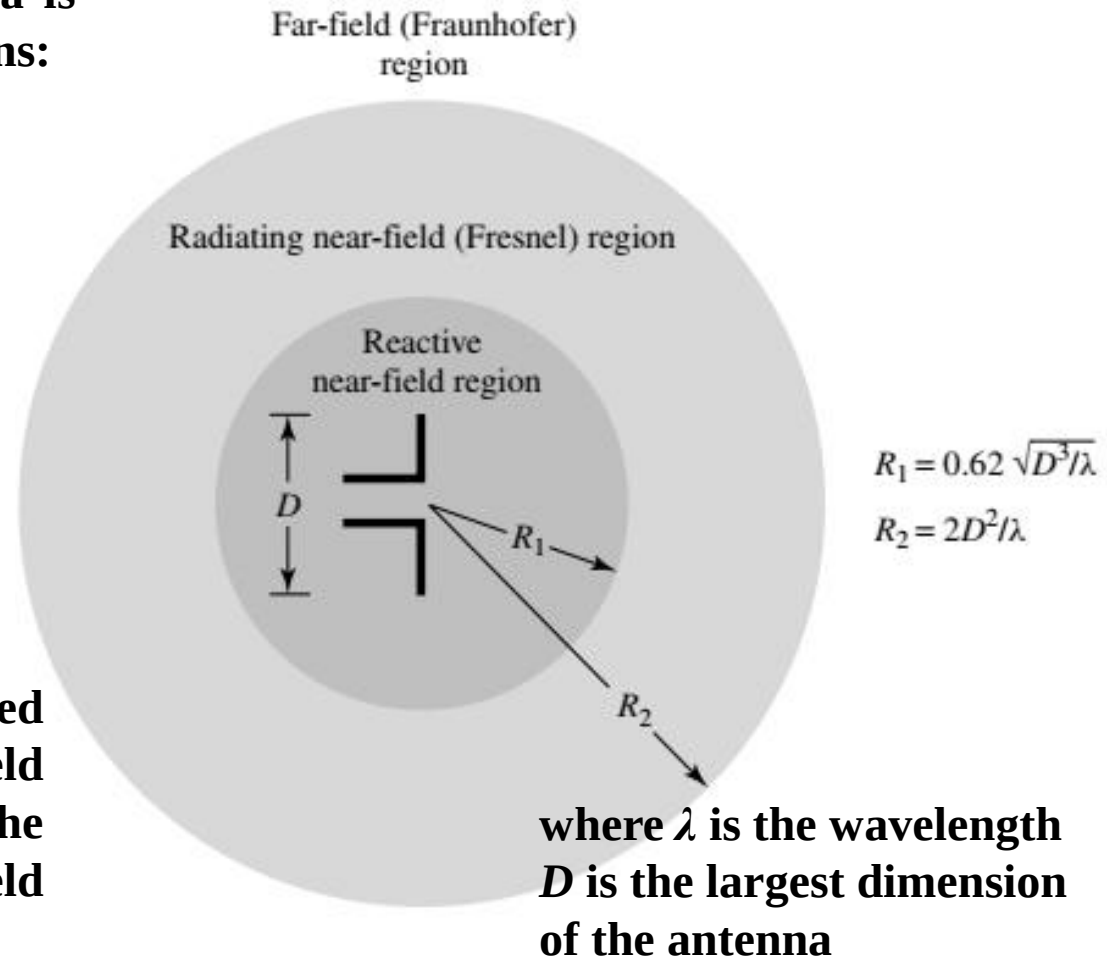


Omnidirectional, pattern have an essentially non-directional pattern in a given plane (in this case in azimuth) and a directional pattern in any orthogonal plane (in this case in elevation).” An *omnidirectional* pattern is then a special type of a *directional* pattern.

Field Regions

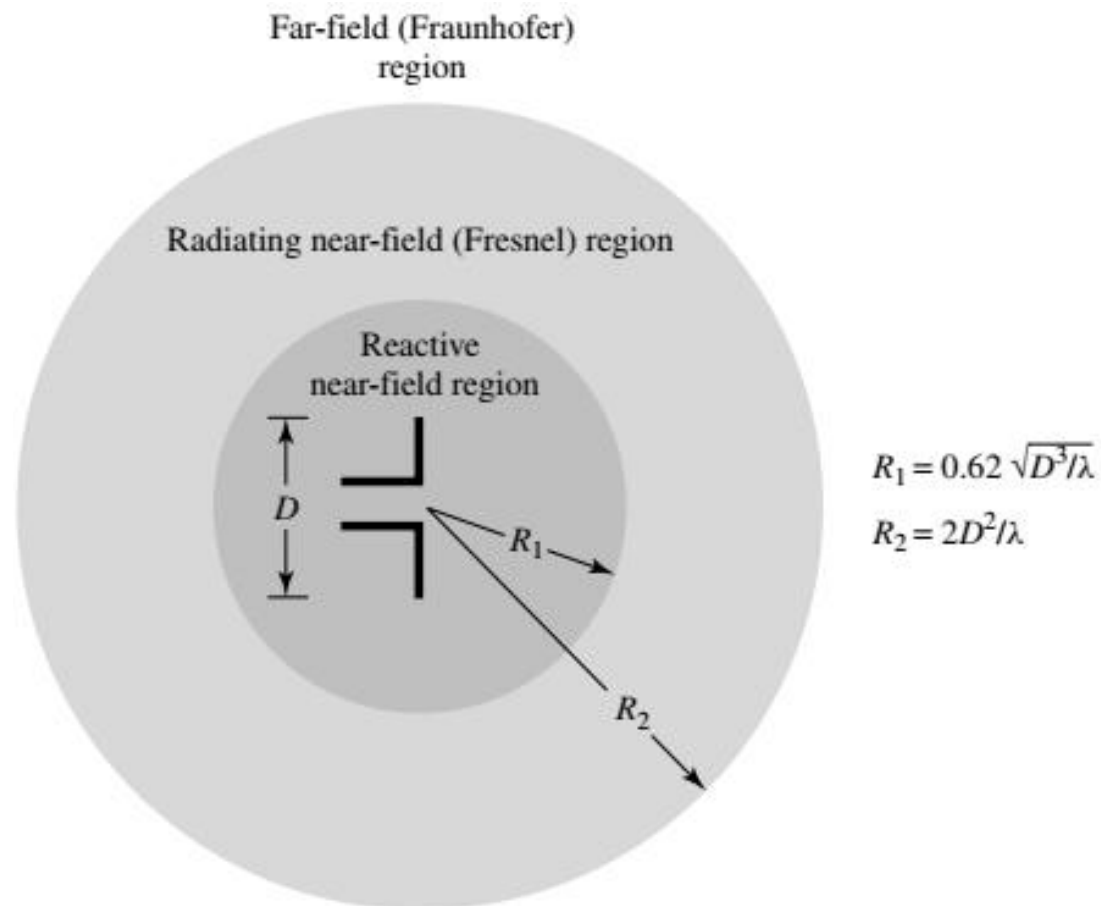
The space surrounding an antenna is usually subdivided into three regions:

- (a) reactive near-field,
- (b) radiating near-field (Fresnel),
- (c) far-field (Fraunhofer) regions

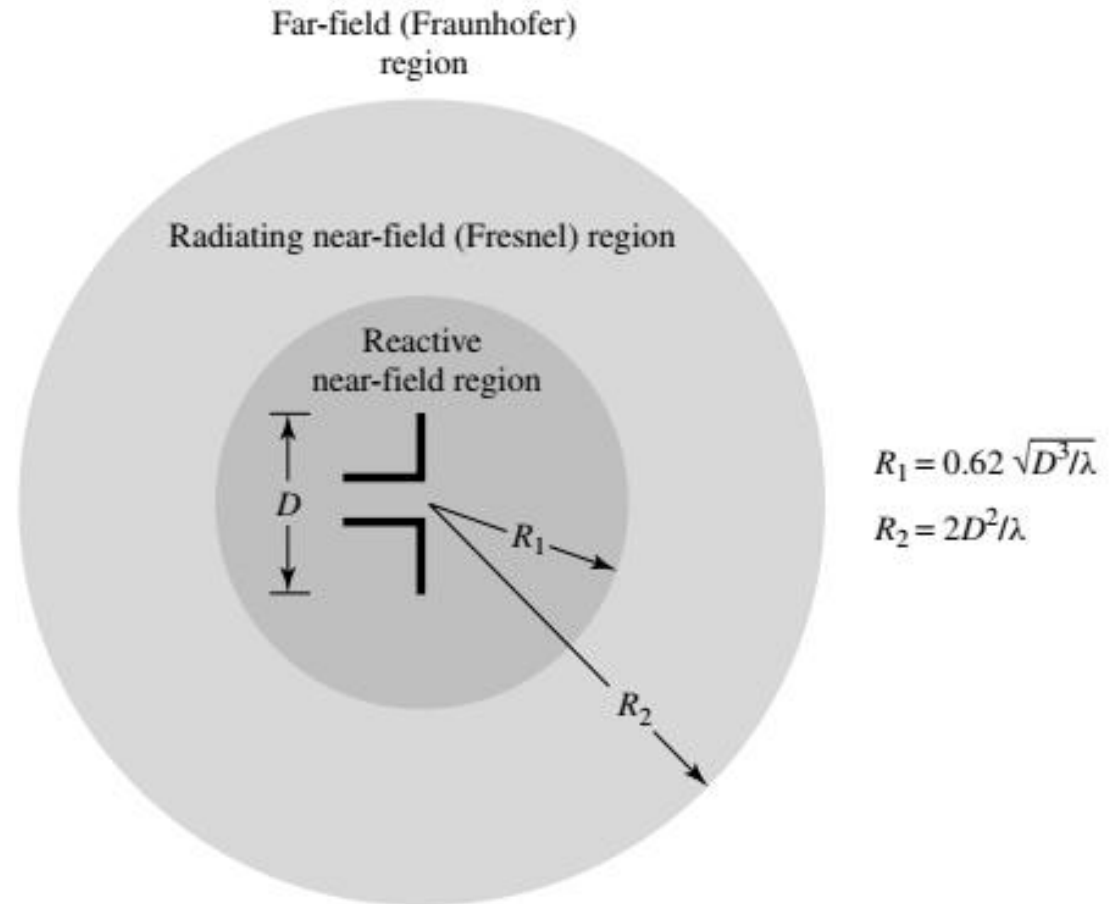


Reactive near-field region is defined as “that portion of the near-field region immediately surrounding the antenna wherein the reactive field predominates.”

Radiating near-field (Fresnel) region is defined as “that region of the field of an antenna between the reactive near-field region and the far-field region wherein radiation fields predominate and wherein the angular field distribution is dependent upon the distance from the antenna.



Far-field (Fraunhofer) region is defined as “that region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna.



RADIATION POWER DENSITY

Electromagnetic waves are used to transport information through a wireless medium or a guiding structure, from one point to the other. It is then natural to assume that power and energy are associated with electromagnetic fields. The quantity used to describe the power associated with an electromagnetic wave is the instantaneous Poynting vector defined as

$$\mathcal{W} = \mathcal{E} \times \mathcal{H}$$

$\mathcal{W}$ = instantaneous Poynting vector (W/m²)

$\mathcal{E}$ = instantaneous electric-field intensity (V/m)

$\mathcal{H}$ = instantaneous magnetic-field intensity (A/m)

$$\mathcal{W} = \mathcal{E} \times \mathcal{H} = \frac{1}{2} \text{Re}[\mathbf{E} \times \mathbf{H}^*] \quad \text{radiation density.}$$

The time average Poynting vector (average power density) can be written as

$$\mathbf{W}_{\text{av}}(x, y, z) = [\mathcal{W}(x, y, z; t)]_{\text{av}} = \frac{1}{2} \text{Re}[\mathbf{E} \times \mathbf{H}^*]$$

(W/m²)

Since the Poynting vector is a power density, the total power crossing a closed surface can be obtained by integrating the normal component of the Poynting vector over the entire surface. In equation form

$$\mathcal{P} = \oiint_S \mathbf{W} \cdot d\mathbf{s} = \oiint_S \mathbf{W} \cdot \hat{\mathbf{n}} da$$

$\mathcal{P}$ = instantaneous total power (W)

$\hat{\mathbf{n}}$ = unit vector normal to the surface

da = infinitesimal area of the closed surface (m^2)

$$\begin{aligned} P_{\text{rad}} = P_{\text{av}} &= \oiint_S \mathbf{W}_{\text{rad}} \cdot d\mathbf{s} = \oiint_S \mathbf{W}_{\text{av}} \cdot \hat{\mathbf{n}} da \\ &= \frac{1}{2} \oiint_S \text{Re}(\mathbf{E} \times \mathbf{H}^*) \cdot d\mathbf{s} \end{aligned}$$

Radiation intensity in a given direction is defined as “the power radiated from an antenna per unit solid angle.” The radiation intensity is a far-field parameter, and it can be obtained by simply multiplying the radiation density by the square of the distance.

The radiation intensity is a far-field parameter, and it can be obtained by simply multiplying the radiation density by the square of the distance. In mathematical form, it is expressed as

$$U = r^2 W_{\text{rad}}$$



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Directivity



Directivity of an antenna is defined as “the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions.

$$D = \frac{U}{U_0} = \frac{4\pi U}{P_{\text{rad}}}$$

$$D_{\text{max}} = D_0 = \frac{U|_{\text{max}}}{U_0} = \frac{U_{\text{max}}}{U_0} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}}$$

As an illustration, find the maximum directivity of the antenna whose radiation intensity is that of Example 2.2. Write an expression for the directivity as a function of the directional angles θ and ϕ .

$$\mathbf{W}_{\text{rad}} = \hat{\mathbf{a}}_r W_r = \hat{\mathbf{a}}_r A_0 \frac{\sin \theta}{r^2} \quad (\text{W/m}^2)$$

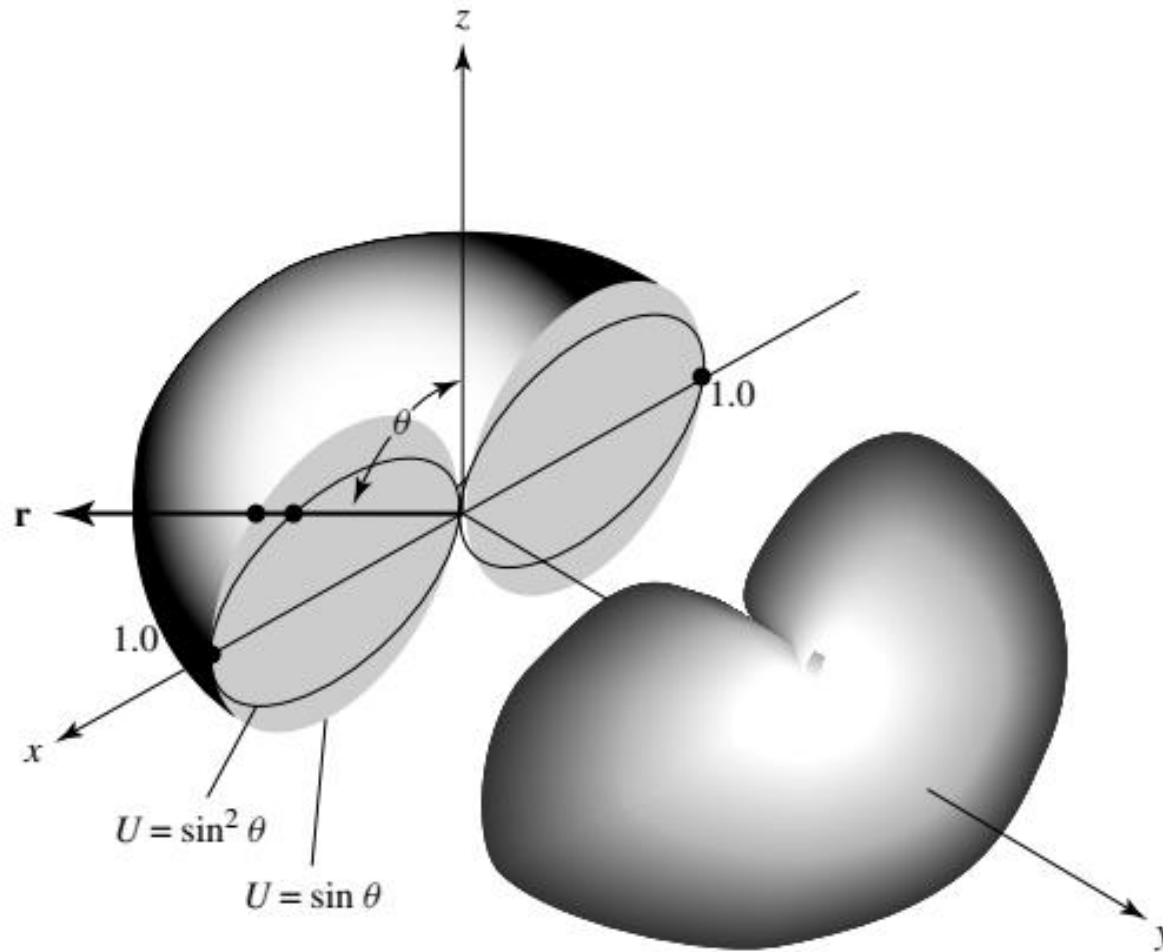


The radial component of the radiated power density of an infinitesimal linear dipole of length $l \ll \lambda$ is given by

$$\mathbf{W}_{\text{av}} = \hat{\mathbf{a}}_r W_r = \hat{\mathbf{a}}_r A_0 \frac{\sin^2 \theta}{r^2} \quad (\text{W/m}^2)$$

where A_0 is the peak value of the power density, θ is the usual spherical coordinate, and $\hat{\mathbf{a}}_r$ is the radial unit vector. Determine the maximum directivity of the antenna and express the directivity as a function of the directional angles θ and ϕ .



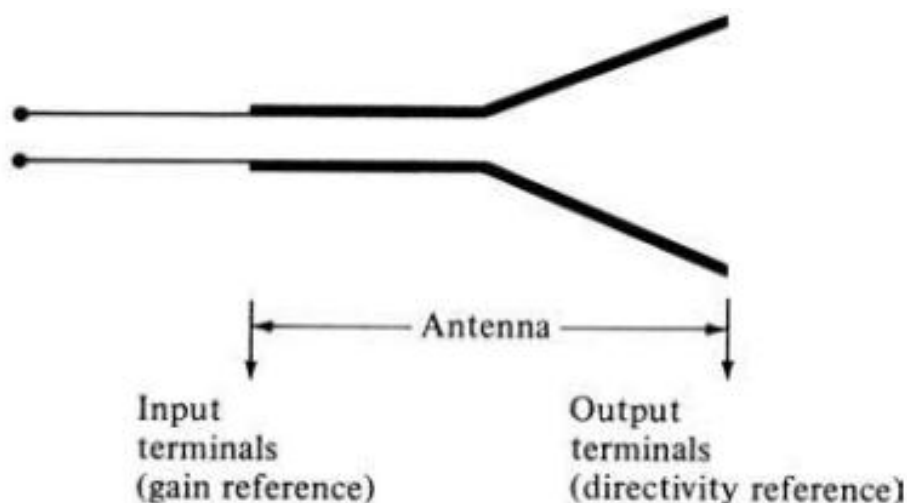


Design an antenna with omnidirectional amplitude pattern with a half-power beamwidth of 90° . Express its radiation intensity by $U = \sin^n \theta$. Determine the value of n and attempt to identify elements that exhibit such a pattern. Determine the directivity of the antenna using

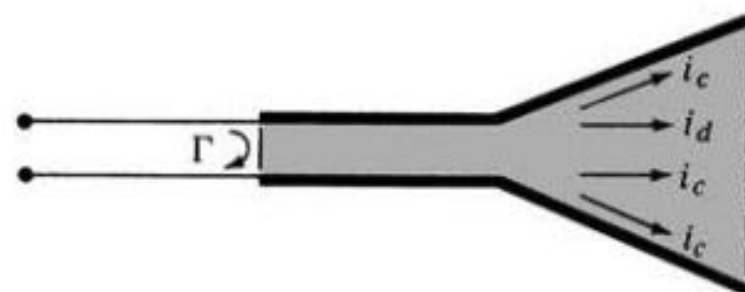
2.8 ANTENNA EFFICIENCY

Associated with an antenna are a number of efficiencies and can be defined using Figure 2.22. The total antenna efficiency e_0 is used to take into account losses at the input terminals and within the structure of the antenna. Such losses may be due, referring to Figure 2.22(b), to

1. reflections because of the mismatch between the transmission line and the antenna
2. I^2R losses (conduction and dielectric)



(a) Antenna reference terminals



(b) Reflection, conduction, and dielectric losses



In general, the overall efficiency can be written as

$$e_0 = e_r e_c e_d$$

where

e_0 = total efficiency (dimensionless)

e_r = reflection (mismatch) efficiency $= (1 - |\Gamma|^2)$ (dimensionless)

e_c = conduction efficiency (dimensionless)

e_d = dielectric efficiency (dimensionless)

Γ = voltage reflection coefficient at the input terminals of the antenna

$[\Gamma = (Z_{in} - Z_0)/(Z_{in} + Z_0)$ where Z_{in} = antenna input impedance,

Z_0 = characteristic impedance of the transmission line]

$$\text{VSWR} = \text{voltage standing wave ratio} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Another useful measure describing the performance of an antenna is the *gain*. *Gain* of an antenna (in a given direction) is defined as “the ratio of the intensity, in a given direction, to the radiation intensity that would be obtained if the power accepted by the antenna were radiated isotropically.

$$\text{Gain} = 4\pi \frac{\text{radiation intensity}}{\text{total input (accepted) power}} = 4\pi \frac{U(\theta, \phi)}{P_{in}} \quad (\text{dimensionless})$$

$$P_{\text{rad}} = e_{cd} P_{in}$$

$$G(\theta, \phi) = e_{cd} \left[4\pi \frac{U(\theta, \phi)}{P_{\text{rad}}} \right] \quad \boxed{G(\theta, \phi) = e_{cd} D(\theta, \phi)}$$